SAI RAKESH ALLA

Electronics & Communication Engineer | VLSI Physical Design | AI & Full-Stack Development

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PROFESSIONAL PROFILE

Electronics and Communication Engineer with industry experience at Electronics Corporation of India Limited (ECIL) and practical training in ASIC Physical Design, AI and Full-Stack Development. Hands-on exposure to RTL-to-GDSII physical design, floorplanning, power planning, placement, clock tree synthesis, routing, timing closure and Static Timing Analysis using Cadence and Synopsys EDA tools. Experienced in TCL scripting, C++, Python, system-level testing, signal validation, EMI/EMC testing and hardware-software integration. Currently pursuing an MSc in Artificial Intelligence with Industrial Practice at the University of Surrey.

PROFESSIONAL EXPERIENCE

Apprentice Engineer | *Electronics Corporation of India Limited (ECIL) — Hyderabad, India* | *Jan 2025 – Mar 2026*

- Executed system-level test activities for the Window Announcement System, analysing signal data and troubleshooting hardware-software integration faults.
- Performed PLC I/O verification and signal validation, identifying anomalies and supporting corrective actions.
- Conducted EMI/EMC testing and interpreted quantitative measurement data against applicable industry requirements.
- Supported PCB, PLC and VME/universal-module activities, system integration, troubleshooting and technical documentation.
- Maintained quality procedure records and documentation in an ISO-certified engineering environment.

PROFESSIONAL TRAINEE

AI & Full-Stack Development Trainee | *MasterNext Academy* | *Apr 2026 – Jul 2026*

- Studied AI foundations, model types and prompt engineering patterns for reliable LLM outputs.
- Built and deployed AI-powered web applications using Lovable and Bolt, progressing from rapid prototypes to functional applications.
- Integrated AI APIs with web interfaces and applied reliability and safety techniques to improve output quality.
- Developed practical skills in HTML, CSS, JavaScript, TypeScript, React and Node.js, with Git/GitHub and modern deployment workflows.

Projects

1. **https://animatedmoviebyrakesh.vercel.app/ - (animatedmovie)**
2. **https://viteandreactbyrakesh.vercel.app/ - (Anatomy of a Web)**
2. **https://elegantlightsbyrakesh.vercel.app/ - (elegantlights)**
3. **https://interiorsbyrakesh.vercel.app/ (interiors)**
4. https://lovable.dev/projects/74235e94-5873-46f9-901d-9f251630b0a6 - (calming meditation and wellness)
5. **https://lovable.dev/projects/f988ab45-984d-4f9d-9a19-d45dec0fa698** -(business data)
6. https://lovable.dev/projects/9a9e6be4-239f-42f2-b396-fa8747308b2e - (Share whatever's on your mind: chatbot)
7. **https://ai.studio/apps/08df5265-902e-40ba-858f-6ae8a5250e1b** - (Moksha Yatra)

Physical Design Engineer Trainee | *VLSI Guru* | *Jun 2024 – Mar 2025*

- Worked through the ASIC Physical Design flow from RTL to GDSII, including floorplanning, power planning, placement, CTS, routing, timing closure and sign-off checks.
- Used Static Timing Analysis concepts to analyse timing and support timing-closure activities.
- Used TCL scripting to automate repetitive design tasks and improve workflow efficiency.
- Developed practical familiarity with Cadence Innovus, Synopsys ICC2 and Synopsys PrimeTime.

KEY PROJECTS

Surveillance Car with Object Detection & Tracking | IoT / AI Project | 2023

- Developed an autonomous surveillance vehicle integrating real-time object detection using OpenCV and Python with line tracking and obstacle avoidance; implemented computer-vision pipelines connecting AI/ML with sensor-driven hardware.

CNTFET-Based Low-Power Ternary CAM | B.Tech Final Year Project | 2023–2024

- Designed a low-power ternary content-addressable memory using Carbon Nanotube FET technology and conducted quantitative performance analysis.

Passport Verification Using RFID | Embedded Systems Project | 2022

- Designed an RFID-based passport verification system with secure data handling and cryptographic data-integrity considerations.

TECHNICAL SKILLS

VLSI / ASIC: ASIC Physical Design, RTL-to-GDSII, Floorplanning, Power Planning, Placement, Clock Tree Synthesis (CTS), Routing, Timing Closure, Static Timing Analysis (STA), Sign-off Checks
EDA Tools: Cadence Innovus, Synopsys ICC2, Synopsys PrimeTime
Programming / Scripting: Python, C++, Embedded C, TCL, JavaScript, TypeScript
AI / ML: Machine Learning, Prompt Engineering, AI API Integration, Computer Vision, OpenCV, LLM Reliability & Safety
Hardware / Embedded: PLC, RFID, IoT, EMI/EMC Testing, Signal Validation, PCB/System Testing
Software / Web: HTML, CSS, React (learning), Node.js (learning), REST APIs, Git, GitHub
Mathematics / Analysis: Probability Theory & Stochastic Processes, Statistics, Numerical Analysis, Signal Processing, Data Structures & Algorithms
Operating Systems: Linux, Windows

EDUCATION

MSc Artificial Intelligence with Industrial Practice | University of Surrey | 2026 – Present
B.Tech Electronics & Communication Engineering | Vignan Foundation Science & Technology Research University | 2020 – 2024 | CGPA: 7.92/10 (First Class)
Diploma Electronics & Communication Engineering | Bapatla Engineering College | 2017 – 2020 | CGPA: 8.78/10

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- Cisco CCNA Certification — Networking Fundamentals and Infrastructure
- IEEE Certification — 5G Wireless Technology
- NPTEL Certification — Joy of Python (Programming Fundamentals)
- NPTEL Certification — Cloud Computing
- Workshop — Ethical Hacking with Cyber Security (TECHNO WORKSHOP)
- Workshop — Xilinx Software Technology (FPGA/EDA Tools)